

Project Development and Completion Tracker

Project ID:	LYAIA610	Class	LY AIA	Percentage of Project Completion:	100%	Copyright Status:	In Progress	Technology Transfer:	NO	
Project Domain (Don't Add New)	Intelligence (AI) & Machine Learning	Source of Problem Statement	Hackathon Problems	Map Sustainable Development Goal: (Don't add New)	18 Peace, Justice and Strong Institutions	Paper Publication Status:	Published	Achievements:	Type Your Answer_1. 2.	
Enter Problem Statement	The growing challenge of detecting manipulated deepfake images that threaten authenticity, security, trust in digital media, and pose risks to society at large.					GitHub Link:			Meeting Count 7	
Project Development Student Team Members	Enrollment Number	Name of Studnets	Class	Contact Number	Email ID	Tech Skill 1	Tech Skill 2	Tech Skill 3	Global Certification 1	Global Certification 2
	MITU22BTC50093	AKSHAY SHARAD JADHAV	LY-AIA-6	shayajadhav2003@gmail.com	9373723606	end Dev-React,Electron,Tail	Java Dev(Basics)	aries, OpenCV,Kera	BM ML from Coursera	
	MITU22BTC50943	UNMESH PRAMOD KAKUSTE	LY-AIA-6	uk21kakuste@gmail.com	9579305872					
	MITU22BTC50506	OM VIJAY MANGATE	LY-AIA-6	vijaymangate1713@gmail.com	9117131713	FrontEnd Dev	Nodejs , Spring Framework	aries, OpenCV,Kera	BM ML from Coursera	
	MITU22BTC50750	SATYAM TANAJI SHINDE	LY-AIA-6	satyamshinde56@gmail.com	9529902376					
Project Guide Industry Mentor	Prof. Sonali Deshpande									

	Review Date	Comment by Jury/ Mentor	Review Link		Technology Stack	1	2	3	4	5
Review-1	19-08-2025				Front-end (Client-side):	React.js	HTML	CSS	JavaScript	Bootstrap
Review-2					Back-end (Server-side):	Flask	FastAPI	Python	TensorFlow Serving	Node.js
Review-3					Database Layer:	MongoDB	Firebase	PostgreSQL	SQLite	MySQL
					Development Tools:	Jupyter Notebook	VS Code	Google Colab	GitHub	Anaconda
					Deployment and Infrastructure:	AWS EC2	Render	Heroku	Docker	GitHub Actions
					Specialization Inlinement:	Computer Vision	Deep Learning	Ensemble Learning	CNN Models	Grad-CAM Visualization

Epics:	As a digital platform user, I should be able to verify the authenticity of uploaded media and detect whether a video is real or fake.
Story 1:	As a user, I should be able to upload an image or video for verification.
Story 2:	As a user, I should get an analysis report showing whether the media is real or deepfake.
Story 3:	As a user, I should be able to understand the detection process through visual explanations (e.g., Grad-CAM heatmaps).



Acceptance Criteria	
1	Users can upload valid images or videos via the web interface.
2	The system accurately classifies uploaded media as real or fake with confidence scores.
3	Grad-CAM visualizations highlight manipulated regions clearly.
4	Ensemble model integrates predictions seamlessly from multiple CNN-The deployed system responds efficiently with minimal latency.

This is actual Development planning and tracking. Don't add documentation, survey, topic finalization, etc. activities.

Sprint No.	Task Name	Sub Tasks	Task Status	Assigned To	Assigned Date	Deadline	Start Date	Completion Date	Completion Status
Sprint 1	Image Processing and Data Acquisition	Image Capture/Acquisition	Complete	Satyam Shinde	8/21/2025	8/29/2025	8/18/2025	8/20/2025	100%
18th Aug 2025		Preprocessing and Enhancement	Complete	Akshay Jadhav	8/25/2025	8/30/2025	8/19/2025	8/22/2025	100%
29th Aug 2025	Data Acquisition & Preprocessing	Feature Extraction	Complete	Unmesh Kakuste	9/1/2025	8/31/2025	8/20/2025	8/22/2025	100%
		Data Storage and Management	Complete	Umesh Kakuste, Satyam Shinde	9/8/2025	9/1/2025	8/21/2025	8/23/2025	100%
Sprint 2	Model Selection & Training	Implement CNN for image-level detection	Complete	Umesh Jadhav, Om Mangrulkar	9/15/2025	2025-09-22	9/15/2025	2025-09-22	100%
1st Sep 2025		Implement RNN/LSTM for video frame temporal analysis	Complete	Umesh Jadhav, Om Mangrulkar	9/22/2025	2025-09-29	9/22/2025	2025-09-29	100%
14th Sep 2025		Implement Autoencoder/GAN-based classifier	Complete	Umesh Jadhav, Om Mangrulkar	9/29/2025	2025-10-06	9/29/2025	2025-10-06	100%
		Train and validate models with initial hyperparameters	Complete	Umesh Jadhav, Om Mangrulkar	10/7/2025	2025-10-14	10/7/2025	2025-10-14	100%
Sprint 3	Model Optimization & Evaluation	Fine-tune hyperparameters (learning rate, batch size, implement cross-model ensemble or voting)	Complete	Umesh Jadhav, Om Mangrulkar	9/15/2025	2025-09-22	9/15/2025	2025-09-22	100%
22nd Sep 2025		Evaluate models using metrics (accuracy, F1-score, AUC)	Complete	Umesh Jadhav, Om Mangrulkar	9/22/2025	2025-09-29	9/22/2025	2025-09-29	100%
5th Oct 2025		Generate explainable outputs (heatmaps, confidence scores)	Complete	Umesh Jadhav, Om Mangrulkar	9/29/2025	2025-10-06	9/29/2025	2025-10-06	100%
		Build user interface or API for image/video	Complete	Umesh Jadhav, Om Mangrulkar	10/7/2025	2025-10-14	10/7/2025		100%
Sprint 4	Sprint 4: Deployment & Integration	Integrate real-time detection and reporting	Complete	Umesh Jadhav, Om Mangrulkar	2025-10-30	2025-11-06			100%
6th Oct 2025		Add logging, privacy measures, and security	Complete	Umesh Jadhav, Om Mangrulkar	2025-11-01	2025-11-07			100%
17th Oct 2025		Test end-to-end system and prepare for release	Complete	Umesh Jadhav, Om Mangrulkar	2025-11-02	2025-11-09			100%

Publication Details									
Sr. No.	Paper Title	Name of Journal	Year	Authors	URL	DOI	Volume	Page no.	Publisher
1	CNN Ensemble Model for	International Journal of Inn	2025	Akshay Jadhav Om Ma	https://ojs.ijournal.in/article/view/7896	4-11-2025	Volume 12 / Issue 6		Editor-in-Chief
2									
Patent Details									
Sr. No.	Title	Inventors	Application No.	Patent Number	Filing Country	Subject Category	Filing Date	Publication Date	Publication Status
1									
Copyright Details									
Sr. No.	Title of work	Name of Applicants	Registration No.	Dairy Number	Date	Status			
1									
Event and Participations Details									
Sr. No.	Name of Event	Type of Event	Date	Type of Participation	Details of Prize won				
1									
2									



Add Weekly Meetings Details [Minimum 12 meetings]							
Meeting No.	08/23/2025	Attendees with commas	ge Capture/Acquisi	Action Items		Assigned to	us (click if completed)
1	08/30/2025	Akshay Jadhav Ummesh Kaku	Preprocessing and	Sprint1.1	Sprint 1.1	@mangateom04@gmail.	✓
2	09/06/2025	Akshay Jadhav Ummesh Kaku	Feature Extraction	Sprint1.3	Sprint 1.2	Ummesh Kaku	✓
3	09/13/2025	Akshay Jadhav Ummesh Kaku	Data Storage and M	Sprint1.4	Sprint 1.3	Om Akshay, Satyam,Un	✓
4	09/20/2025	Akshay Jadhav Ummesh Kaku	Implementation CNN for	Sprint1.5	Sprint 1.3	Om Akshay, Satyam,Un	✓
5	09/27/2025	Akshay Jadhav Ummesh Kaku	Implement RNN.LS	Sprint1.6	Sprint 1.4	Om Akshay, Satyam,Un	✓
6	10/01/2025	Akshay Jadhav Ummesh Kaku	Implement RNN.LS	Sprint1.7	Sprint 1.5	Om Akshay, Satyam,Un	✓
7	10/11/2025	Akshay Jadhav Ummesh Kaku	Implement RNN.LS	Sprint1.8	Sprint2.6	Om Akshay, Satyam,Un	✓

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9							☐		
10							☐		
11							☐		
12							☐		